

Customer Churn Prediction Using Machine Learning

Executive Report

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1. Executive Summary

Customer churn remains one of the most significant challenges facing subscription-based businesses, particularly in the telecommunications industry. Accurately identifying customers who are likely to leave enables organizations to implement targeted retention strategies, improve customer satisfaction, and reduce revenue loss.

This project developed and evaluated three machine learning classification models to predict customer churn using customer demographic information, subscribed services, contract details, billing information, and payment methods. The models were compared using multiple evaluation metrics, including Accuracy, Precision, Recall, F1-score, Confusion Matrix, and ROC-AUC.

Among the evaluated models, Logistic Regression achieved the best overall performance with an accuracy of 80.5% and the highest F1-score, making it the recommended model for deployment.

2. Business Problem

Customer acquisition is considerably more expensive than customer retention. Understanding which customers are most likely to churn allows organizations to proactively engage high-risk customers before they leave.

The primary objective of this project was to develop a predictive model capable of identifying customers likely to churn while also uncovering the key factors driving customer attrition.

3. Dataset Overview

Dataset: IBM Telco Customer Churn Dataset

Observations: 7,032 customers

Target Variable: Churn

Problem Type: Binary Classification

The dataset contains customer demographic information, subscribed services, contract type, payment methods, billing information, tenure, and churn status.

4. Exploratory Data Analysis

The exploratory analysis revealed several important patterns associated with customer churn:

- Customers on Month-to-Month contracts experienced significantly higher churn rates than those on One-Year or Two-Year contracts.
- Customers with higher Monthly Charges were more likely to churn.
- Customers with shorter tenure demonstrated substantially higher attrition.
- Customers using Electronic Check as a payment method showed elevated churn rates.
- Customers subscribed to Online Security and Tech Support services exhibited lower churn rates.

These findings provided valuable insights into customer behavior and informed subsequent predictive modeling.

5. Machine Learning Models

Three supervised machine learning models were developed and evaluated:

- Logistic Regression
- Decision Tree
- Random Forest

Model performance was assessed using Accuracy, Precision, Recall, F1-score, Confusion Matrix, and ROC-AUC.

6. Model Performance

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	80.5%	65.0%	57.2%	60.9%
Decision Tree	77.8%	58.1%	59.6%	58.8%
Random Forest	79.1%	64.4%	47.9%	54.9%

Logistic Regression demonstrated the strongest overall performance, achieving the highest Accuracy and F1-score while maintaining a balanced trade-off between Precision and Recall.

7. Key Business Insights

The analysis identified several factors strongly associated with customer churn:

Factors Increasing Churn Risk

- Month-to-Month contracts

- Higher Monthly Charges
- Electronic Check payment method
- Fiber Optic Internet service

Factors Reducing Churn Risk

- Longer customer tenure
- One-Year and Two-Year contracts
- Online Security service
- Tech Support service
- Customers with dependents

These insights can help organizations prioritize retention efforts and improve customer loyalty.

8. Recommendations

Based on the findings, the following recommendations are proposed:

- Encourage customers to migrate to longer-term contracts through targeted incentives.
- Strengthen customer engagement during the early stages of the customer lifecycle.
- Identify high-risk customers using the predictive model and implement proactive retention campaigns.
- Promote value-added services such as Online Security and Tech Support.
- Monitor customers with high monthly charges and provide personalized offers where appropriate.

9. Conclusion

This project successfully developed and evaluated multiple machine learning models for customer churn prediction. Logistic Regression emerged as the best-performing model, achieving an accuracy of 80.5% and the highest F1-score among the evaluated algorithms.

Beyond predictive performance, the project generated actionable business insights into the primary drivers of customer churn. These findings can support data-driven customer retention strategies, improve customer satisfaction, and reduce revenue loss.

Figure 1. Comparison of Machine Learning Model Performance

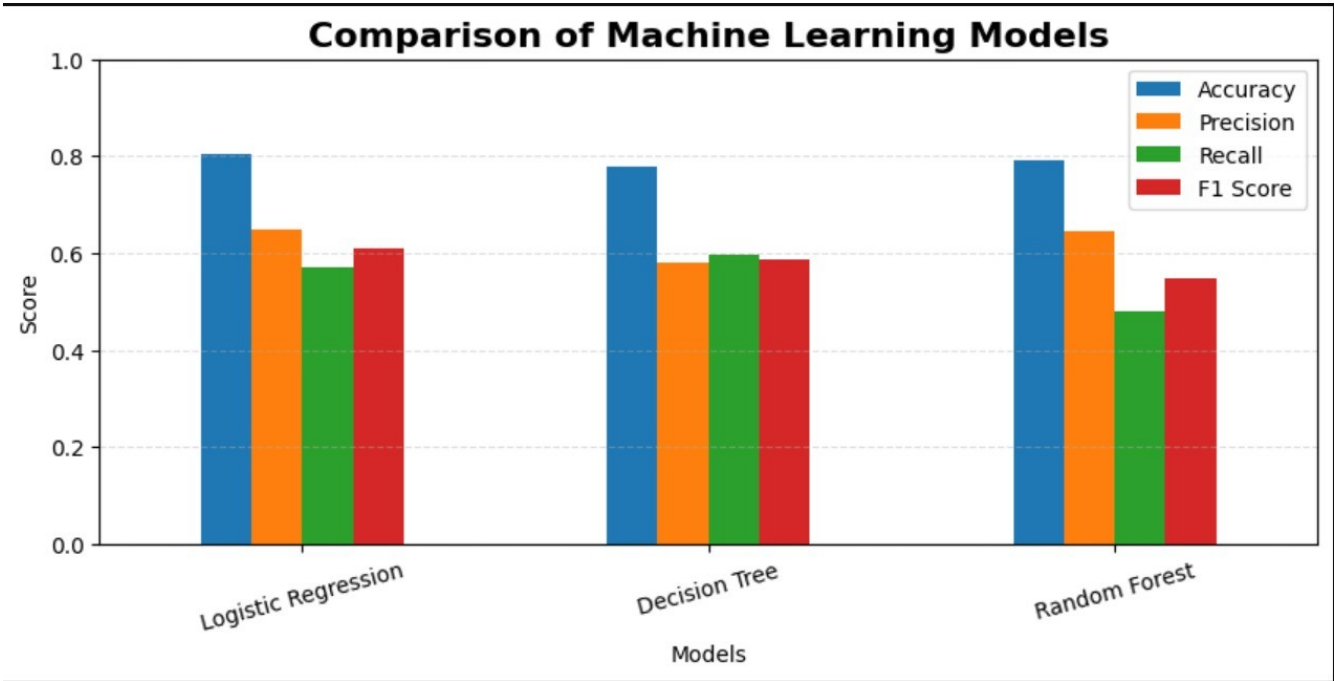


Figure 2. ROC Curve

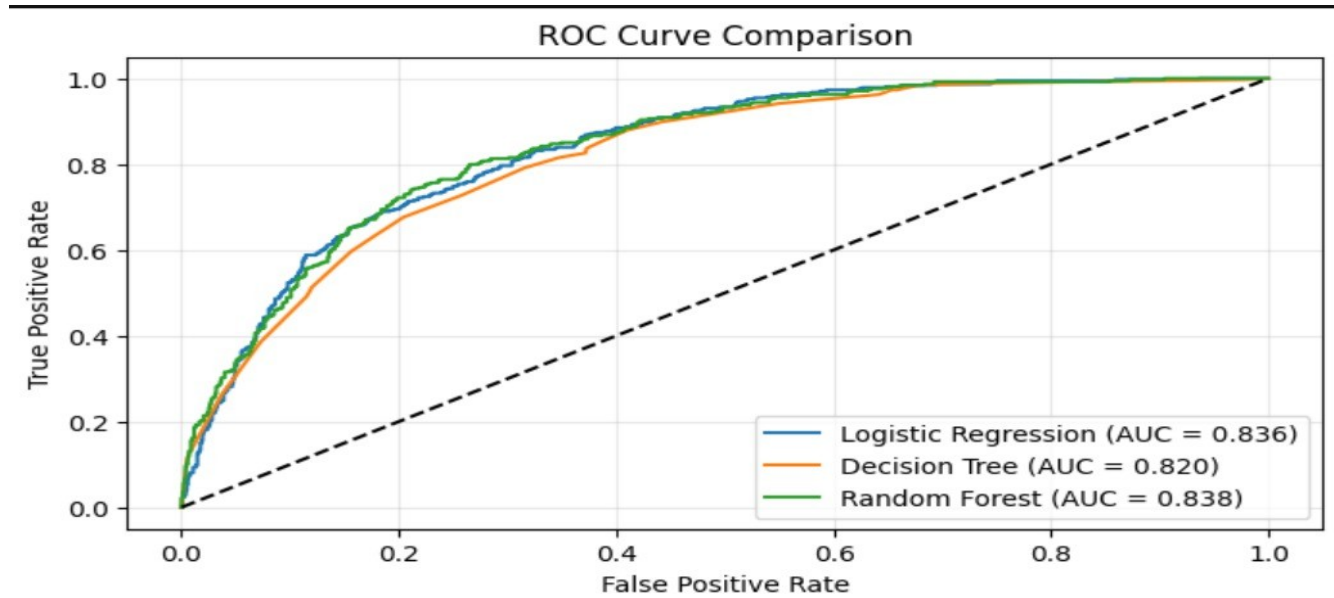


Figure 3. Confusion Matrix

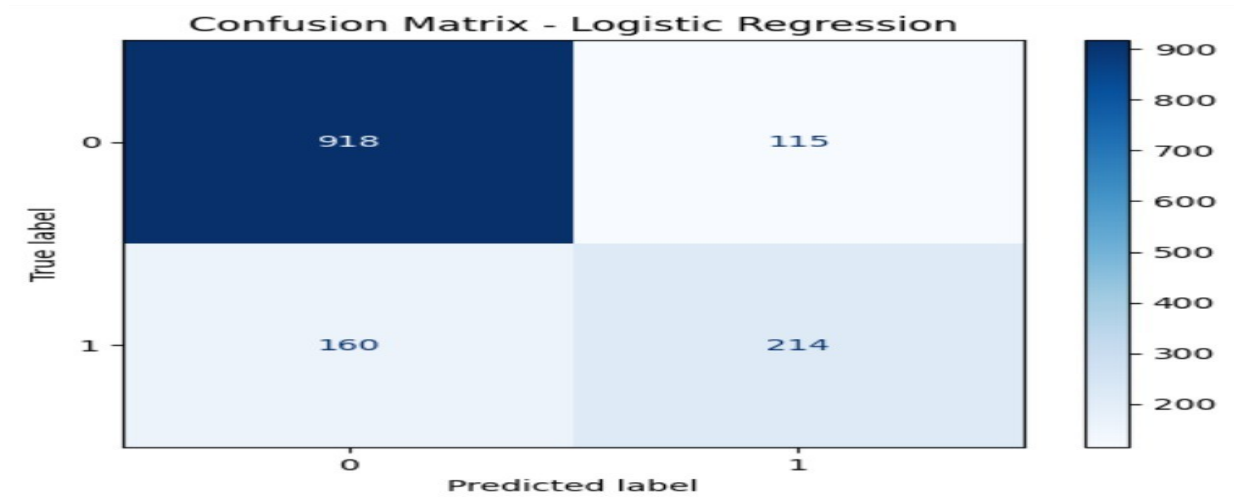
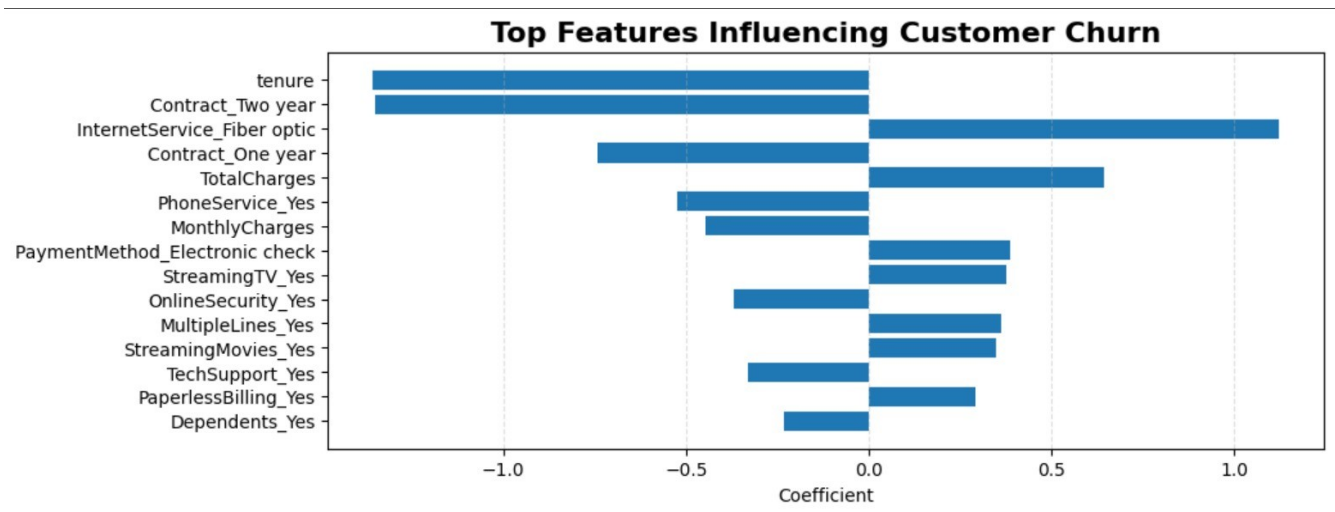


Figure 4. Feature Importance of Plot



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